

Module - 3

Correlation

Definition:

- * Correlation analysis deals with the association b/w two or more variables.
- * Correlation analysis attempts to determine the degree of relationship b/w variables.
- * Correlation is an analysis of the covariation b/w two or more variables.

Types of correlation



positive and negative correlation.

Whether correlation is +ve or -ve, would depend upon the direction of the change of the variables.

Correlation is +ve when the values of the two variables move in the same $\rightarrow$ so that an $\uparrow$ in the value of one variable is followed by an $\uparrow$ in the value of the other variable.

(or) a $\downarrow$ in the value of one variable is followed by a $\downarrow$ in " the other variable -

(e.g) i) amount of rainfall and yield of crops.

ii) price and demand of a commodity.

Correlation is said to be -ve, when the values of the two variables move in the opposite $\rightarrow$ so that an $\uparrow$ in the value of one variable is followed by a $\downarrow$ " of the other variable.

(e.g) i) vaccination and illness

ii) Sales of woolen garments and temperature.

Simple correlation

When we consider only two variables and check the correlation b/w them is said to be a simple correlation.

Multiple correlation

When we consider 3 or more variables for correlation simultaneously..

(e.g.) yield of rice, amount of rainfall and amount of fertilizers used.

Partial correlation

When one or more variables are kept constant and the relationship b/w others, is termed as partial correlation.

Methods of studying correlation

Graphical

↓
Scatter diagram.

Algebraic method

✓
Karl-pearson's
co-eff. of
correlation

✓
Spearman's
rank correl.
co-eff.

Concurrent
deviating
method

Scatter diagrams

- is the simplest method of studying correlation b/w two variables.

* collect n -pairs of values of x and y correct.

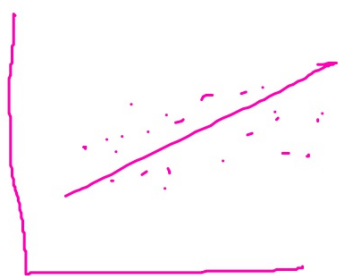
to n -repetitions of a random experiment.

Let them be $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, which can be plotted in a graph, called Scatter diagram, using which we can form a fairly good though vague idea of relationship b/w x and y .

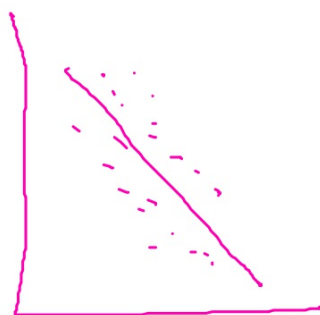
* If the points are dense or closely packed then x and y are correlated.

on the other hand, if the points are widely scattered throughout the graph paper, then x and y are either not correlated or poorly correlated.

Further if the points in a scatter diagram appear to lie near a straight line, then we can assume that the RV's have linear correlation:



positive correlation



negative correlation



no correlation

In this section, we assume that the RV's are linearly correlated and discuss how to measure the degree of linear correlation.

Karl-pearson's product moment correlation w-eff.

(or) Karl Pearson's correlation co-eff.

$$* \text{Cov}(X, Y) = E\{[X - E(X)][Y - E(Y)]\}$$

is called product moment of X and Y .

* Karl Pearson's correlation co-eff. b/w X & Y is given by

$$r(X, Y) = r_{xy} = \frac{\text{Cov}(X, Y)}{\sigma_x \cdot \sigma_y}$$

(i.e.)

$$r_{xy} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2} \sqrt{\sum (y - \bar{y})^2}}$$

properties of correlation co-eff.

① $-1 \leq r_{xy} \leq 1$

-1	0	1

* when $0 < r_{xy} \leq 1$, then the correlation b/w x and y is positive or direct.

* $-1 \leq r_{xy} < 0$, then correlation is negative or inverse

② Two independent RV's x and y are uncorrelated but two correlated RV's need not be independent.

③ compute the co-eff of correlation b/w x and y using the following data.

$x :$	1	3	5	7	8	10
$y :$	8	12	15	17	18	20.